

Salem Lahlou

Assistant Professor of Machine Learning

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Research interests : AI for science tools, generative flow networks, LLM reasoning.

Experience

- Sep 2024 – **Assistant Professor of Machine Learning**, MBZUAI, Abu Dhabi.
- 2024 **Senior Researcher**, Technology Innovation Institute, Abu Dhabi.
- 2018 – 2023 **Researcher**, Mila, Montréal - Canada.
- 2017 **Data Scientist**, Booking.com, Amsterdam.

Internships

- 2020 **Researcher**, Google Research, Paris.
- 2015 **Researcher**, IBM Research, Singapore.
- 2014 **Data Scientist**, Infosys, Bangalore.

Education

- 2018 – 2023 Mila — **Université de Montréal**, Montréal - Canada, PhD, Supervisor : Yoshua Bengio.
- 2015 – 2016 **École Normale Supérieure**, Cachan - France, M.Sc..
- 2012 – 2016 **École Polytechnique**, Palaiseau - France, B.Sc. + M.Sc..

Research grants (awarded)

- Jul 2026 **IAAI Research Program grant : “Trustworthy Agronomic Reasoning for Smallholder Advisory” (24 months)**, PI; Co-PIs : Martin Takáč, Nils Lukas, 500,000 USD over 24 months.
- Apr 2026 **Grant from MBZUAI for research on advancing GFlowNet research**, PI, Co-PI : Nils Lukas, 500,000 AED.
- Jun 2025 **Etihad flagship research project (24 months)**, Co-PI, PI : Nils Lukas, 450,000 USD per year.
- Jan 2025 **EnergyAI flagship research project (30 months)**, Co-PI, PI : Martin Takáč, 2M USD per year.

Selected publications

(* : equal contribution)

- 2026 **PROBE : Learning to Audit Policy Compliance in Tool-Using LLM Agents**, Kshitij Mishra, Abhijith Sharma, Nils Lukas, **Salem Lahlou**. NeurIPS 2026.
- 2026 **Expert-Aided Causal Discovery of Ancestral Graphs**, Tiago da Silva, Bruna Bazaluk, Eliezer de Souza da Silva, António Góis, **Salem Lahlou**, Dominik Heider, Samuel Kaski, Diego Mesquita, Adèle Helena Ribeiro. Information Sciences.
- 2026 **Particle GFlowNets : Rethinking Generative Marginalization Models**, Tiago Silva, Diego Mesquita, **Salem Lahlou**. UAI 2026.
- 2026 **Path-dependent Discrete Amortized Inference (Oral)**, Tiago Silva, Nikolay Malkin, **Salem Lahlou**. ICML 2026.
- 2026 **Noise-Corrected GRPO : From Noisy Rewards to Unbiased Gradients**, Omar El Mansouri, Fathinah Asma Izzati, Mohamed El Amine Seddik, **Salem Lahlou**. ICML 2026.
- 2026 **CoRe : Collaborative Reasoning via Cross Teaching**, Kshitij Mishra, Mirat Aubakirov, Martin Takáč, Nils Lukas, **Salem Lahlou**. ICML 2026.
- 2026 **FinChain : A Symbolic Benchmark for Verifiable Chain-of-Thought Financial Reasoning**, Zhuohan Xie, Dhruv Sahnun, Debopriyo Banerjee, Georgi Georgiev, Rushil Thareja, Hachem Madmoun, Jinyan Su, Aaryamonvikram Singh, Yuxia Wang, Rui Xing, Fajri Koto, Haonan Li, Ivan Koychev, Tanmoy Chakraborty, **Salem Lahlou**, Veselin Stoyanov, Preslav Nakov. ACL 2026.
- 2026 **Temporal Contrastive Decoding : A Training-Free Method for Large Audio-Language Models**, Yanda Li, Yuhua Liu, Zirui Song, Yunchao Wei, Martin Takáč, **Salem Lahlou**. Findings of ACL 2026.

- 2026 **Byzantine Cheap Talk : Adversarial Resilience and Topology Effects in LLM Coordination Games**, Aya El Mir, Martin Takáč, **Salem Lahlou**. *NETYS 2026*.
- 2026 **Communication Enables Cooperation in LLM Agents : A Comparison with Curriculum-Based Approaches**, Hachem Madmoun, **Salem Lahlou**. *EACL 2026 (Oral)*.
- 2026 **TorchGFN : A PyTorch GFlowNet Library**, Joseph D. Viviano*, Omar G. Younis*, Sanghyeok Choi*, Victor Schmidt, Yoshua Bengio, **Salem Lahlou**. *JMLR 2026(+ NeurIPS 2025 Workshop on Frontiers in Probabilistic Inference)*.
- 2025 **Loss-guided Auxiliary Agents for Overcoming Mode Collapse in GFlowNets**, Idriss Malek, Aya Laajil, Abhijith Sharma, Eric Moulines, **Salem Lahlou**. *AAAI 2026*.
- 2025 **Improved Exploration in GFlowNets via Enhanced Epistemic Neural Networks**, Sajan Muhammad, **Salem Lahlou**. *ICoIAS 2025*.
- 2025 **PORT : Preference Optimization on Reasoning Traces**, **Salem Lahlou***, Abdalgader Abubaker*, Hakim Hacid. *NAACL 2025*.
- 2025 **On the Privacy Risks of Spiking Neural Networks : A Membership Inference Analysis**, Junyi Guan*, Abhijith Sharma*, Chong Tian, **Salem Lahlou**. *UAI 2025 (+ oral at ICLR 2025 Workshop ICBINB)*.
- 2025 **SVRPBench : A Realistic Benchmark for Stochastic Vehicle Routing Problem**, Ahmed Heakl*, Yahia Salaheldin Shaaban*, Martin Takac, **Salem Lahlou**, Zangir Iklassov. *NeurIPS 2025*.
- 2023 **A Theory of Continuous Generative Flow Networks**, **Salem Lahlou**, Tristan Deleu, Pablo Lemos, Dinghuai Zhang, Alexandra Volokhova, Alex Hernández-García, Léna Néhale Ezzine, Yoshua Bengio, Nikolay Malkin. *ICML 2023*.
- 2023 **GFlowNet Foundations**, Yoshua Bengio*, **Salem Lahlou***, Tristan Deleu*, Edward J. Hu, Mo Tiwari, Emmanuel Bengio. *JMLR 2023*.
- 2023 **GFlowNets and Variational Inference**, Nikolay Malkin*, **Salem Lahlou***, Tristan Deleu*, Xu Ji, Edward Hu, Katie Everett, Dinghuai Zhang, Yoshua Bengio. *ICLR 2023*.
- 2023 **DEUP : Direct Epistemic Uncertainty Prediction**, **Salem Lahlou***, Moksh Jain*, Hadi Nekoei, Victor Butoi, Paul Bertin, Jarrid Rector-Brooks, Maksym Korablyov, Yoshua Bengio. *TMLR 2023*.
- 2019 **BabyAI : A Platform to Study the Sample Efficiency of Grounded Language Learning**, Maxime Chevalier-Boisvert*, Dzmitry Bahdanau*, **Salem Lahlou**, Lucas Willems, Chitwan Saharia, Thien Huu Nguyen, Yoshua Bengio. *ICLR 2019*.

Service (selected)

- 2026–2027 **ML Program Admissions Lead**, Graduate (MSc and PhD) admissions for the Machine Learning program, MBZUAI CMS division.
- 2026–2027 **Area Chair**, *ICLR 2026 and 2027, ICML 2026, NeurIPS 2026, AAAI 2027 (Senior PC)*.
- 2026 **Local Chair**, *AISTATS 2026*.
- 2025–2026 **Program Chair / PC Co-Chair**, *NETYS 2025 ; WebConf 2026 PhD Symposium ; FAISys 2026 (Xiamen, December)*.
- 2026 **Main Organizer**, *MBZUAI ML Winter School 2026*.
- 2022–2026 **Reviewing**, *ICLR, ICML, NeurIPS, UAI, AAAI, ECAI, TMLR, IEEE TPAMI, JASA, ACL, NETYS, and more*.
- 2025–2026 **MBZUAI committees**, *Faculty Hiring, Admissions (ML + Undergraduate), ML Curriculum redesign (Jan–May 2025), TA Chair (Jan–Oct 2025)*.
- 2025–2026 **Jury member**, *1 PhD Defense jury ; 2 PhD Candidacy juries ; 12 MSc Thesis juries*.